Félix Wilhelmy

Machine Learning Engineer — Deep Learning & Applied AI

Education

Master's Degree with Thesis 2024–2026
École de Technologie Supérieure
Quantum Machine Learning

Graduate Coursework 2025
MILA / Université de Montréal
Representation Learning (IFT6135)

AlgoLab Summer School 2024
Institut Quantique / Université de Sherbrooke
Near-Term Quantum Algorithms

Bachelor of Engineering 2018–2023
École de Technologie Supérieure
Software Engineering

DEC 2013–2016
Collège Montmorency
Computer Science

Awards

1st Place, iQuCodeFest 2025
Polytechnique Montréal
Quantum Computing Hackathon

2nd Place, YQuantum 2025
Yale Quantum Institute
Quantum Optimization Hackathon

Best Effort Award, Mil'HaQ Fest 2025
MILA
Quantum Machine Learning Hackathon

Languages

French (Native)
English (Professional Proficiency)

Contact

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felix-wilhelmy
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About

Software engineer who has just graduated with a master's degree in quantum machine learning, a cumulative GPA of 4.06/4.30 (A), and an Excellent distinction for my thesis defense. Persistent, rigorous, and deeply committed to the projects I undertake, I combine hands-on industry experience in software development with a strong research background in machine learning.

My experience has led me to solve complex technical problems, develop and optimize machine learning models, and design and deploy high-performance software tools. From scientific experimentation to real-world systems, I thrive in demanding environments where I value teamwork and adapt quickly to new challenges. A scientific publication based on my master's research is currently in preparation.

Projects

QB-PIER – Adversarial Robustness of Classical, Quantum, and Hybrid Models 2024–2026
Master's Research Project - École de technologie supérieure

- Built a platform to compare how image classifiers withstand adversarial attacks
- Solely developed the reproducible Python pipeline from scratch to train and evaluate 37+ classical, quantum, and hybrid models, using FGSM and black-box transfer attacks

Adaptive Graph Capture Strategy 2026
MIT iQuHack 2026 (competition) - Cambridge, MA, USA

- Developed a strategy to capture the most useful nodes in a competitive graph with limited resources
- Combined adaptive Bell-pair entanglement-distillation circuits with classical graph heuristics and budget safeguards

Photonic LSTM for Swaption Volatility 2025
Mil'HaQ Fest 2025 - Mila / Université de Montréal

- Developed a model to forecast changes in financial-market volatility from historical data
- Adapted an LSTM for photonic quantum circuits, forecasting each tenor–maturity pair locally on a 3×3 swaption grid

Resource Optimization for Wildfire Response 2025
Yale YQuantum 2025 (competition) - New Haven, CT, USA

- Developed a method to direct limited wildfire-response resources to the highest-risk areas
- Processed GeoTIFF risk maps with geographic cropping and downsampling, then formulated the allocation as a QUBO and applied QAOA

FR2SQL – French Text-to-SQL 2025
MTI820 (Academic Project) - École de Technologie Supérieure

- Designed a French-to-SQL prototype around LLaMA 3 8B and QLoRA

Skills

Programming

Python, SQL, C++, C#, TypeScript, data structures, algorithms

Deep Learning

CNNs, ResNets, MLP-Mixers, recurrent networks, transformers, VAEs, diffusion models

Applied AI

PyTorch, PennyLane, Hydra, TensorBoard, computer vision, NLP, adversarial robustness, data analysis

Engineering Tools

Git, Docker, DuckDB, Streamlit, CI/CD (Jenkins), data pipelines, experiment tracking

Experience

Esports Data Analyst May 2026–Sep 2026
Canadian National Team - Overwatch World Cup 2026

- Developed a Python/DuckDB system processing an estimated 180 000 in-game events per practice day to generate metrics, dashboards, and reports
- Analyzed player, team, and opponent performance to support coaches' preparation and competitive decisions

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President (24–25) & VP Competitions (23–24) Sep 2023–Jan 2026
QuantumÉTS (Student Quantum Computing Club) - Montréal, QC

- Led technical teams in machine learning and quantum computing competitions, from project definition through experimentation and delivery
- Mentored students on algorithmic problem solving, experimental implementation, and technical communication, contributing to several podium results in international hackathons

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Teaching Assistant Jan 2024–Aug 2025
École de Technologie Supérieure - Montréal, QC

- Supported teaching in quantum computing and machine learning by delivering lectures and assisting students with practical coursework.

Full Stack Developer (Internship) Sep 2022–May 2023
Amilia - Montréal, QC

- Developed and maintained production features using C# and Angular in a large-scale SaaS application
- Improved Jenkins CI/CD workflows and supported production software through testing, debugging, issue investigation, and technical documentation

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Esports Coach 2016–2026
International Competitive Teams - International

- Led and strategically coached internationally recognized competitive teams, applying leadership, critical analysis, and decision-making under pressure

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Research Assistant (Internship) Apr 2021–Aug 2021
LASSENA Research Laboratory - Montréal, QC

- Developed C++/Qt software for integrating and analyzing aeronautical navigation instruments in a research environment
- Implemented UDP-based communication and data-handling components under real-time processing constraints

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Full Stack Developer (Internship) May 2019–Sep 2019
Air Transat - Montréal, QC

- Developed an internal flight-request management application using C#, Angular, TypeScript, and SQL
- Migrated the frontend from Angular 7.2 to Angular 8 and optimized a critical SQL query, achieving an approximately sixfold performance improvement

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Research & Development (Internship) Mar 2016–May 2016
National Film Board of Canada - Montréal, QC

- Built Python tools to train and evaluate a Tesseract OCR system, designing experiments and test workflows for applied ML research